TALHA RANA

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PROFESSIONAL SUMMARY

AI Backend Engineer with 3+ years of experience architecting and deploying production-grade LLM agents, RAG pipelines, and cloud-native APIs using Python, FastAPI, LangChain, LangGraph, and AWS. Experienced in remote-first delivery, async collaboration, and independently shipping end-to-end AI systems. Two systems currently live in production: a therapy-automation pipeline that has processed 500+ sessions and a HIPAA-scoped legal platform with 20+ users.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Bash

AI / LLMOps: LangChain, LangGraph, LangSmith, vLLM, Qdrant, Guardrails AI, ElevenLabs STT/TTS, Whisper, StyleTTS2, Claude, Gemini, OpenAI, LLaMA

Backend: FastAPI, Flask, REST API Design, Microservices, JWT Auth, WebSockets, Tool Calling / Function Calling

Frontend: Next.js, React Native (Expo), ElectronJS, TypeScript

Databases/Cache: PostgreSQL, MongoDB, MySQL, Redis, Valkey, SQLAlchemy / SQLModel, Alembic

Cloud & DevOps: AWS (EC2, S3, SES, SQS, ALB, Elastic Beanstalk, Amplify, CloudFront, RDS, KMS, WAF, Secrets Manager), Docker, GitHub Actions, Cloudflare R2, Sentry

Testing & Tooling: pytest, Unit Testing, LangSmith Tracing, Selenium, Playwright

PROFESSIONAL EXPERIENCE

AI Backend Engineer | Kodexo Labs — Apr 2025 – Present | Hybrid | Karachi, Pakistan

- Built modular LLM agent pipelines with LangChain and LangGraph, traced and debugged with LangSmith.
- Integrated Guardrails AI across inference endpoints to enforce structured outputs.
- Built FastAPI inference APIs with tool / function calling for multi-step AI workflows.
- Set up CI/CD with Docker and GitHub Actions for multi-environment AWS deployments (EC2, S3, ALB).
- Built conversational and audio AI workflows with ElevenLabs STT/TTS, Claude Sonnet 4.6 and SendGrid.

Full Stack Engineer | Bleed AI — Sep 2023 – Apr 2025 | Remote | United States

- Integrated StyleTTS2 and Whisper speech features into cross-platform production applications.
- Built and maintained a cross-platform Electron desktop application in production.
- Designed PostgreSQL data models with SQLModel and optimised query plans for production loads.
- Built FastAPI REST APIs with real-time streaming for AI product features.
- Provisioned cloud infrastructure on AWS (EC2, S3) and Cloudflare R2.

PROJECTS

Angel — Therapy Automation Pipeline (live) *FastAPI, MongoDB, ElevenLabs Scribe v2 & v3 Alpha, Claude Sonnet 4.6, AWS*

- Turns therapy session audio into transcripts, summaries, podcast scripts and multi-speaker audio; ~500 sessions processed in production.
- Replaces 4–5 hours of therapist work per session with a 20–30 minute pipeline run, with a therapist approval step before script generation.
- GDPR-compliant, encrypted at rest and in transit; deployed on Elastic Beanstalk with CloudFront, WAF, SQS, KMS and Secrets Manager.

Settle Wing — Legal Services Platform (live) *FastAPI, MongoDB, Stripe, Sentry, Elastic Beanstalk, Amplify, GitHub Actions*

- Owned backend, Stripe payments and AWS infrastructure for a platform handling legal documents including medical records; live with 20+ users.
- HIPAA-scoped encryption across database, cloud, payments and error monitoring; dev, staging and production released through GitHub Actions.

Diesel Repair — Intelligent RAG System *FastAPI, PostgreSQL (RDS), Qdrant, LLaMA 4 Maverick, OpenAI embeddings*

- Owned backend and AI integration for a diagnostic assistant over 15,000+ repair notes, embedded with textembedding-3-small and searched in Qdrant.
- Internal testing traced shallow answers to terse, flat source notes rather than to retrieval

Fanbai — AI Character Chat Platform *FastAPI, vLLM, PostgreSQL, SQLModel, Alembic, Stripe, Cloudflare R2, Next.js*

- Backend for a previously live character-chat product: personalities read from metadata embedded in PNG character cards, replies streamed from a self-hosted model on vLLM.
- Stripe payments, R2 asset storage, Alembic-managed PostgreSQL schema; integrated the API with the Next.js frontend.

Hooked AI — Social Media Growth Platform *FastAPI, LangChain, MongoDB, Stripe, S3*

- Agent-Supervisor architecture: a supervisor agent routes work to a content generation agent and a content growth agent.

Amarco — AI Agent Tool Calling System *FastAPI, LangChain, LangSmith, MongoDB, Microsoft Graph API*

- Agent with tools for email, calendar and SharePoint via Microsoft Graph, with results persisted to MongoDB and traced in LangSmith.

Real-Time Conversational AI (prototype) *FastAPI, Distil-Whisper, OpenAI, StyleTTS2*

- Voice-to-voice STT → LLM → TTS loop built as a client demo, with speech models self-hosted on a local GPU and pause-based turn-taking

Smart Trading Card App — Frontend *Expo React Native, FastAPI, Azure Vision API*

- Built the React Native app: camera and photo-library capture, integrated with an OCR backend for card inventory and valuation.

Academia Navigator — Web Scraping Automation *FastAPI, Selenium, BeautifulSoup4*

- Course-outline extraction with Selenium and BeautifulSoup4 that cut data collection from days to hours.

Thesia Character — AI Desktop Platform *ElectronJS, Next.js, TypeScript, FastAPI, PostgreSQL, Cloudflare R2*

- Cross-platform AI desktop app with a FastAPI / PostgreSQL / R2 backend and oobabooga text-generation integration.

EDUCATION

Bachelor of Science in Computer Science CGPA: 3.6 / 4.0

FAST National University, Karachi, Pakistan | Aug 2019 – May 2023

Languages: English (Advanced C1 / Fluent) • Urdu (Native)